

Lab 8

Arrays and Nested Loops

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Sample Programs

1. Program to copy 1 array to another array.

Note: We have used indexed addressing this time to iterate through array.

```
.data
numbers    DW  34, 45, 56, 67, 75, 89
numbers2   DW  6 DUP (0)                ;initialize numbers2 with 0,0,0,0,0 (6 times)

.code
main proc
    mov ecx, 6                        ;initialize counter to array size
    mov esi, 0                        ;initialize offset to 0

move_next:
    mov ax, [numbers + esi]          ;store value at numbers + offset to eax
    mov [numbers2 + esi], ax         ;store value in ax to numbers2 + offset
    add esi, 2                        ;increment your offset
    loop move_next                   ;loop until ecx == 0

    Invoke ExitProcess, 0

main endp
end main
```

2. Program to find the sum of natural numbers until a number for each given number in the array. These sum results are stored in the sum array in the corresponding indices.

Note: Nested loops are used

```
.data
numbers    DW  6, 8, 9, 5
sum         DW  4 DUP(?)

.code
main proc
    MOV ECX, 4
    MOV ESI, 0
    OUTER_LOOP:
        MOV AX, 0
        MOV BX, [numbers + ESI]
        INNER_LOOP:
            ADD AX, BX
            DEC BX
            CMP BX, 0
            JNZ INNER_LOOP
        MOV [sum+ESI], AX
        ADD ESI, 2
        LOOP OUTER_LOOP
```

Lab Work

- Take reference from the above sample program and implement following programs.
- Attach properly commented code in the lab report.
- Add flow chart of your logic for each program implemented.

Question 1:

You have given an array of numbers. Find square of each number and store it in result array.

.data		
numbers	DW	2, 4, 3, 7
result	DW	4 DUP (0)

Question 2:

You have given an array of numbers. Find factorial of each number and store it in result array.

.data		
numbers	DW	2, 4, 3, 7
result	DW	4 DUP (1)
size	DW	4